Voice Scribe v4 — The Stitch Policy, by Intuition

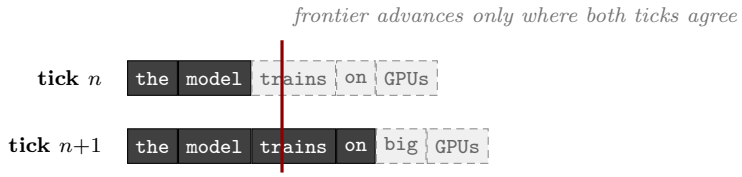
Every tuning parameter of `stream_policy.py`: the failure without it, and what the tuned value buys

2026-08-05 (policy as of the post-review resolution pass; C7.14 + C9.3 included)

Legend. `word` committed (permanent) `word` tentative (may still change) `word` wrong text `word` duplicate | commit frontier

0. The engine

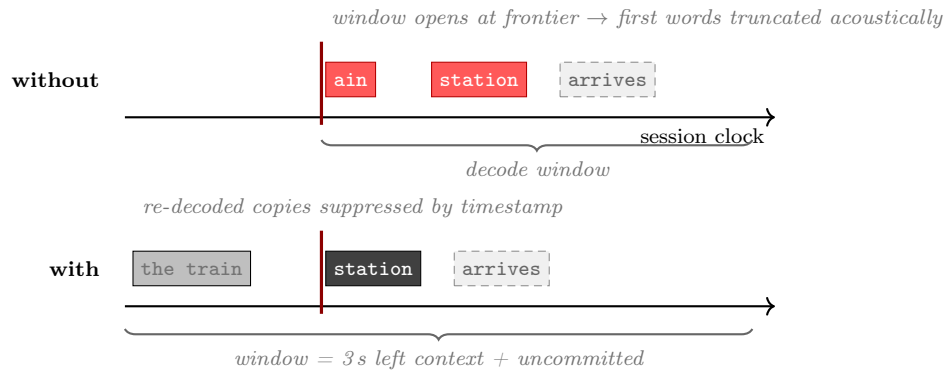
Every 750 ms tick the server re-decodes one window: 3 s of already-committed context plus all uncommitted audio. The policy compares this hypothesis with the previous tick's and commits the longest *agreeing prefix* (case- and punctuation-insensitive). Everything else stays tentative. Structural rule (C5.5): every word either commits or withholds — *none revises*. That is why "zero retractions" is a property of the construction, not a hope.



The hypotheses agreed on `trains on`; those commit. The tail (`big GPUs` vs `GPUs`) disagrees; it waits. Ten knobs bound this loop. Each figure below shows the degenerate setting first — the failure the knob exists to prevent — then the tuned behavior.

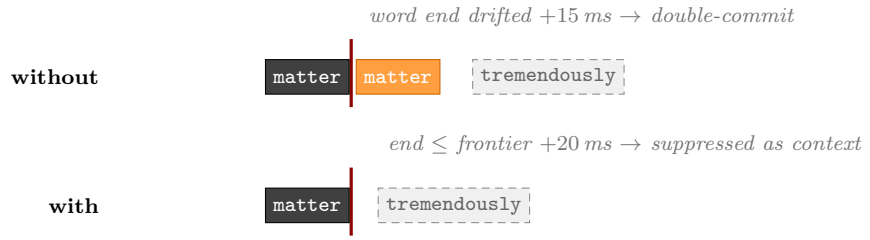
1. `left_context_s = 3.0` — the decoder never starts cold

Without: the window opens exactly at the frontier; the model decodes mid-phrase with no conditioning and mangles the first words. **With:** 3 s of committed audio is re-decoded for conditioning only; `localize()` drops the copies *by timestamp*, never by string match (string matching fails exactly when the model re-punctuates).



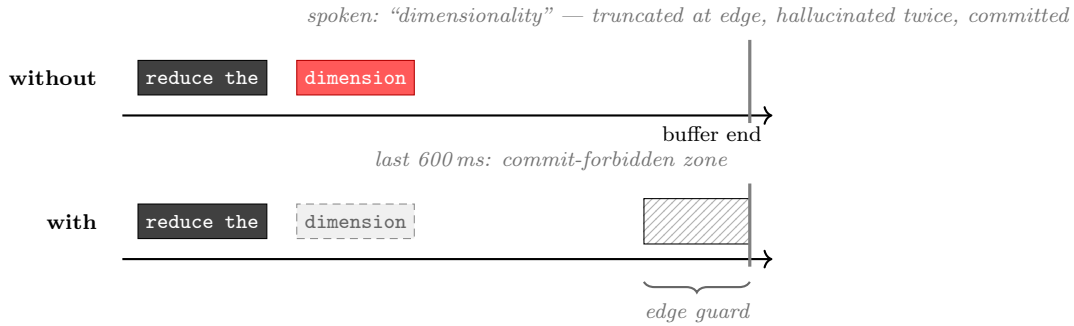
2. `frame_tolerance_s = 0.02` — one decoder frame of drift

Without: a re-decode re-times the last committed word's end 15 ms past the frontier; it survives the boundary filter and commits *again* — append-only means the duplicate is permanent and invisible to the retraction reducer. **With:** anything ending within one Whisper frame (20 ms) of the frontier is treated as re-decoded committed audio.



3. `edge_guard_s = 0.6` — agreement is not enough at the edge

Without: a word cut off at the buffer edge makes the model hallucinate a completion — *the same completion, with good probability, two ticks in a row.* Agreement commits it; the real word later differs. **With:** words ending in the last 600 ms can never commit, agreed or not. This is the structural defense (C7.4): distance from the edge, not confidence, is the test.

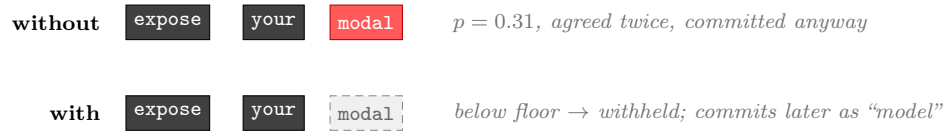


4. `word_prob_floor = 0.5` — a veto, not a rejection

Without: an agreed word at 0.31 probability commits permanently. **With:** the commit scan stops at the first agreed word under the floor; the word stays tentative and re-tries next tick with more

audio behind it. Vetoes only withhold — they never edit — so the retraction guarantee holds.

Field note (2026-08-05 trace diagnosis): the floor's own failure mode is **hostage-taking** — on utterance 16 of the gate fixture, one word at $p \approx 0.3$ held a nine-word agreed prefix behind it for three consecutive ticks, until the holdback valve (§6) burst-committed all ten unverified. One low word can convert a healthy agreement run into a forced commit; the per-tick stall counters (§6) now make this visible per session.



5. The terminal-word rule (structural — no knob)

The last word of a hypothesis never commits: Whisper re-punctuates and re-capitalizes it on nearly every pass, and committed text takes the surface form of the *newer* hypothesis. Field note (2026-08-04 wire log): this is exactly why sentence-final punctuation drops when a commit boundary lands on a sentence end — the period existed only while the word was terminal.

Measured and settled (2026-08-05, offline replay on the gate fixture): committing the *older* surface instead was evaluated against the one-shot reference — 18 divergent commits, prev-correct 11 vs prev-spurious 6. Both hypotheses fabricate punctuation in both directions, and the same word commits *with* its period when the cut lands after the sentence boundary: the defect is cut-position-dependent, not hypothesis-choice-dependent. Current behavior stands; a punctuation-aware hybrid is parked for v4.2.



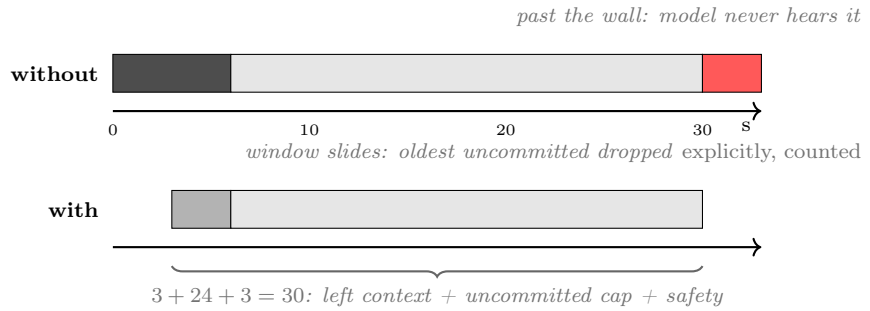
6. `max_holdback_ticks = 6` — the anti-stall valve

Without: disfluent speech makes consecutive hypotheses flip-flop; the agreed prefix stops growing and the frontier stalls *forever* — unbounded commit lag. **With:** after 6 stalled ticks ($6 \times 750\,\text{ms} = 4.5\,\text{s}$) the newest hypothesis force-commits without agreement or the probability floor — but never past the edge guard. The valve's ledger from the 2026-08-04/05 field runs (impromptu speech, clean measurement): `force_commit_rate` 8.7% — how often it fired; `forced_span_wer` 14.3% — the quality price of committing unagreed text; `commit_lag_max` 4.1–6.1 s — the bound at work. The stall-reason counters (C9.3, landed 2026-08-05) decompose *why* the valve gets armed: of 96 zero-commit ticks on the gate fixture, 55 were the terminal guard (agreement reaching only the last word), 20 the probability floor, 18 genuine divergence, 3 the edge guard. The v4.2 tuning question is therefore short-agreement behavior — not generic hypothesis churn.



7. `max_uncommitted_s = 24` + `safety_s = 3` — the 30 s wall

Whisper's input is a fixed 30 s window; `pad_or_trim` *silently truncates* anything longer. The invariant $3 + 24 + 3 = 30$ (enforced in `validate()`) makes silent truncation impossible. **Without:** the uncommitted span grows past the wall and the model simply never hears the overflow — unaccounted audio loss. **With:** `plan_window` reports the overflow and the session drops the *oldest* uncommitted samples explicitly, counted in `dropped_audio_ms`. The 2026-08-04 u3 run slid three 79 s utterances through this and reported `dropped_audio_ms = 0` with zero retractions — the slide worked without ever needing the drop.

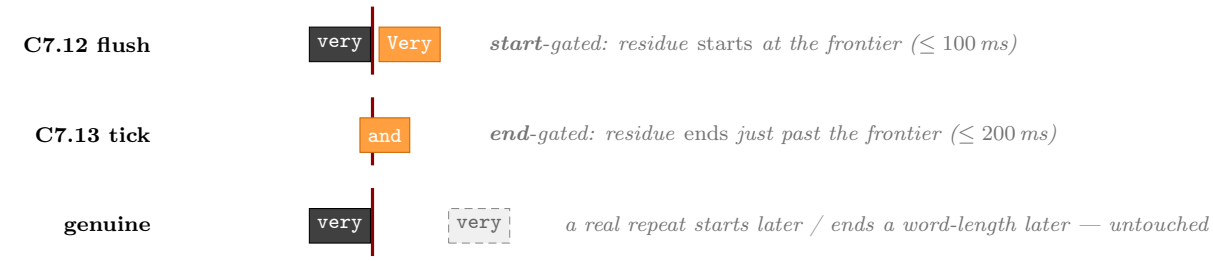


8. `flush_left_context_s = 1.0` — context priced in latency

The flush decode (end of utterance — silence is ground truth, commit everything) opens only 1 s before the frontier, not 3: the flush sits on the finalization-latency path, and *every second of context here is paid directly in the user-visible finalization number* (gate: ≤ 0.5 s after the first utterance; 2026-08-04 measured 0.544 s). `validate()` refuses a flush context larger than `left_context_s` — retention only guarantees that much audio still exists.

9. The twin residue guards — `seam_tolerance_s = 0.10` (C7.12) and `tick_residue_tolerance_s = 0.20` (C7.13)

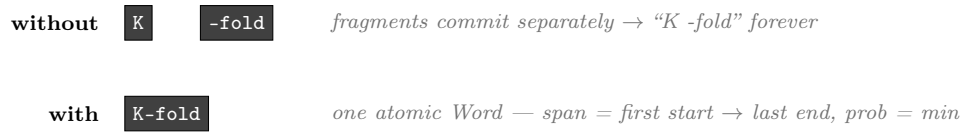
Same disease, two symptoms. The frontier is a DTW word-end *estimate* that lands tens of ms early, so acoustic residue of the last committed word leaks into the next decode and re-emits ("very Very", 2026-08-02 field finding). Both guards are conjunctions on the *leading* word matching the last committed word — but they gate on opposite ends, because the impostors differ:



Accepted residual risk, stated in the code: a genuine immediate repeat falling inside the tolerance is indistinguishable by any local rule and is dropped — so every suppression is counted on the session-close log line, keeping the trade observable. 2026-08-04 field ledger: 3 seam catches, 2–14 tick-residue suppressions per session, zero visible duplicates.

10. `merge_glued_words` — the decoder-edge merge (C7.14, new)

Runs *before* everything above, at the decoder edge. faster-whisper marks word-initial tokens with a leading space and emits glued continuations (`"-fold"`, `"%"`) without one. The policy previously stripped that signal and re-joined committed words with plain spaces — fabricating `"K -fold"` and `"95 %"` (9 artifact tokens per 412 words live, each scoring ≈2 WER errors against the space-preserving one-shot reference). The merge makes each glued cluster *one* Word, so a commit boundary can never split it and every downstream space-join is correct by construction.



Measured effect on the gate fixture: `wer_micro` 4.55% → 3.68% — that share of the "streaming quality gap" was an assembly artifact, not recognition.

Coda — how the knobs met the field on 2026-08-04/05

One recording of disfluent, self-correcting impromptu speech exercised every mechanism above: the holdback valve fired at 8.7% (its ledger, §6), the slide handled 3×79 s utterances without dropping a sample (§7), the twin guards caught frontier residue while one burst-commit landed past the C7.12 start-gate (§9's accepted residual, now diagnosed — see §4's hostage note), the terminal-word rule explained the missing sentence periods (§5, measured and settled), and the flush context closed to 13 ms over its latency gate (§8) once the decoder-edge merge (§10) cleaned the measurement. Every number above is now reproducible offline: the C9.3 trace + `policy_replay.py` re-ran 291 live GPU ticks with zero drift. Tuning any one knob moves a number you can now name — and test in milliseconds.